

# git log/history

## References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

## Git History - Directory Commit Statistics

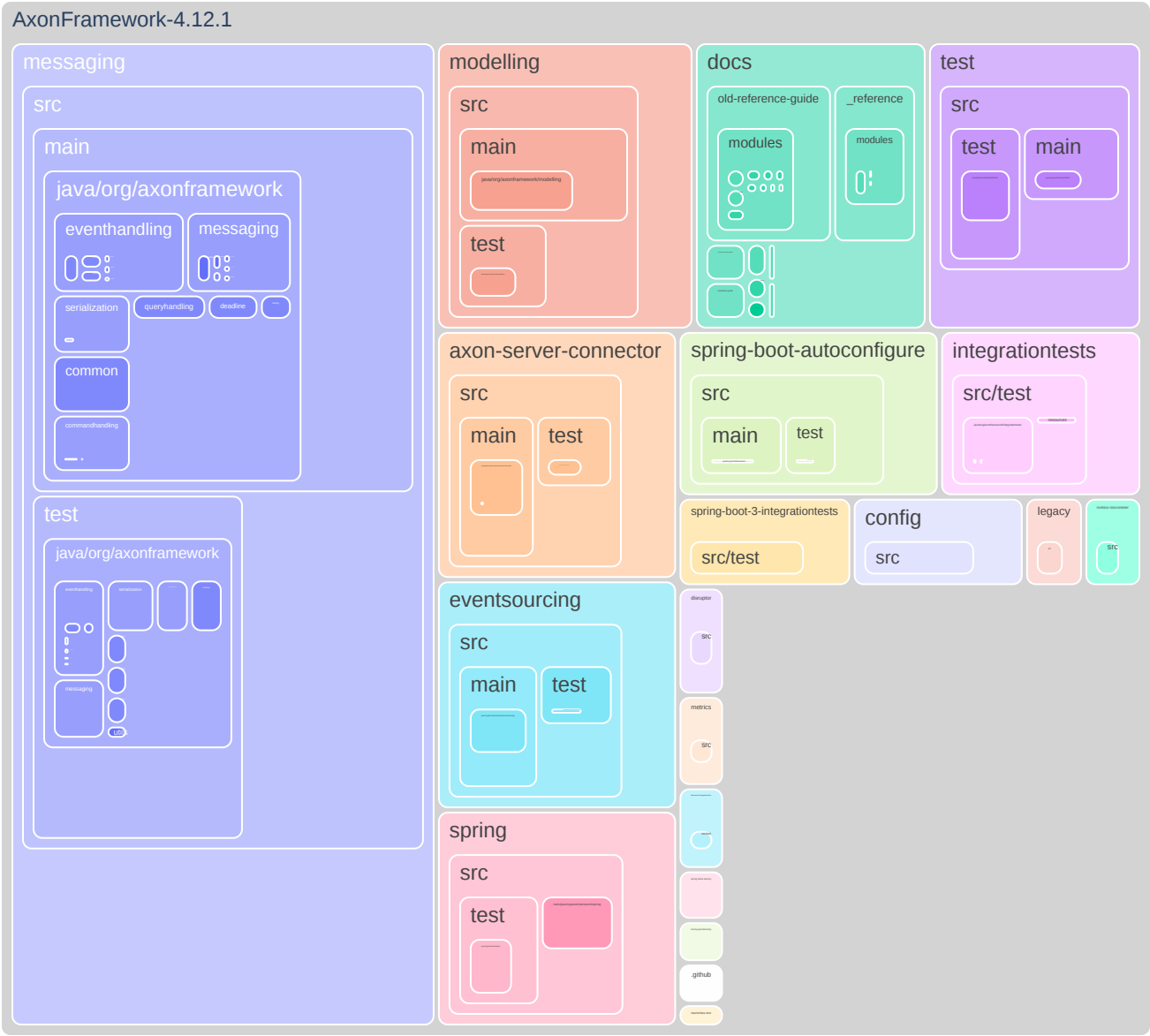
### Data Preview

|       | fileCount   | authorCount | commitCount | daysSinceLastCommit | daysSinceLastCreation | daysSinceLastModification |
|-------|-------------|-------------|-------------|---------------------|-----------------------|---------------------------|
| count | 564.000000  | 564.000000  | 564.000000  | 564.000000          | 564.000000            | 564.000000                |
| mean  | 24.898936   | 19.664894   | 288.570922  | 317.186170          | 1193.590426           | 410.828014                |
| std   | 120.739280  | 21.297173   | 639.823202  | 170.624646          | 980.427734            | 392.532956                |
| min   | 1.000000    | 2.000000    | 3.000000    | 39.000000           | 52.000000             | 38.000000                 |
| 25%   | 1.000000    | 7.000000    | 27.000000   | 179.000000          | 452.000000            | 269.000000                |
| 50%   | 4.000000    | 12.000000   | 82.500000   | 412.000000          | 971.000000            | 411.000000                |
| 75%   | 10.000000   | 25.000000   | 270.250000  | 412.000000          | 2018.000000           | 411.000000                |
| max   | 2294.000000 | 213.000000  | 9815.000000 | 2562.000000         | 5689.000000           | 3040.000000               |

|    | directoryPath                                                                                                    | directoryParentPat                                               |
|----|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 0  | AxonFramework-4.12.1/assembly                                                                                    | AxonFramework-4.12.                                              |
| 1  | AxonFramework-4.12.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source | AxonFramework-4.12.1/axon-serve                                  |
| 2  | AxonFramework-4.12.1/axon-server-connector/src/main/resources/META-INF/services                                  | AxonFramework-4.12.1/axon-server-connector/src/main/resource     |
| 3  | AxonFramework-4.12.1/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util       | AxonFramework-4.12.1/axon-serve                                  |
| 4  | AxonFramework-4.12.1/config/src/test/resources/META-INF                                                          | AxonFramework-4.12.1/config/src/test/resource                    |
| 5  | AxonFramework-4.12.1/coverage-report                                                                             | AxonFramework-4.12.                                              |
| 6  | AxonFramework-4.12.1/docs/_playbook/localLinks                                                                   | AxonFramework-4.12.1/docs/_playboo                               |
| 7  | AxonFramework-4.12.1/docs/_reference/modules/ROOT/attachments                                                    | AxonFramework-4.12.1/docs/_reference/modules/ROO                 |
| 8  | AxonFramework-4.12.1/docs/_reference/modules/ROOT/partials                                                       | AxonFramework-4.12.1/docs/_reference/modules/ROO                 |
| 9  | AxonFramework-4.12.1/docs/_reference/modules/axon-server-connector/pages                                         | AxonFramework-4.12.1/docs/_reference/modules/axon-server-connect |
| 10 | AxonFramework-4.12.1/docs/_reference/modules/axon-server-connector/partials                                      | AxonFramework-4.12.1/docs/_reference/modules/axon-server-connect |
| 11 | AxonFramework-4.12.1/docs/_reference/modules/configuration/pages                                                 | AxonFramework-4.12.1/docs/_reference/modules/configuratio        |
| 12 | AxonFramework-4.12.1/docs/_reference/modules/configuration/partials                                              | AxonFramework-4.12.1/docs/_reference/modules/configuratio        |
| 13 | AxonFramework-4.12.1/docs/_reference/modules/disruptor/pages                                                     | AxonFramework-4.12.1/docs/_reference/modules/disruptor           |
| 14 | AxonFramework-4.12.1/docs/_reference/modules/disruptor/partials                                                  | AxonFramework-4.12.1/docs/_reference/modules/disruptor           |
| 15 | AxonFramework-4.12.1/docs/_reference/modules/eventsourcing/pages                                                 | AxonFramework-4.12.1/docs/_reference/modules/eventssourcin       |
| 16 | AxonFramework-4.12.1/docs/_reference/modules/eventsourcing/partials                                              | AxonFramework-4.12.1/docs/_reference/modules/eventssourcin       |
| 17 | AxonFramework-4.12.1/docs/_reference/modules/legacy/pages                                                        | AxonFramework-4.12.1/docs/_reference/modules/legac               |
| 18 | AxonFramework-4.12.1/docs/_reference/modules/legacy/partials                                                     | AxonFramework-4.12.1/docs/_reference/modules/legac               |
| 19 | AxonFramework-4.12.1/docs/_reference/modules/metrics/pages                                                       | AxonFramework-4.12.1/docs/_reference/modules/metric              |
| 20 | AxonFramework-4.12.1/docs/_reference/modules/metrics/partials                                                    | AxonFramework-4.12.1/docs/_reference/modules/metric              |
| 21 | AxonFramework-4.12.1/docs/_reference/modules/micrometer/pages                                                    | AxonFramework-4.12.1/docs/_reference/modules/micromete           |
| 22 | AxonFramework-4.12.1/docs/_reference/modules/micrometer/partials                                                 | AxonFramework-4.12.1/docs/_reference/modules/micromete           |
| 23 | AxonFramework-4.12.1/docs/_reference/modules/modeling/partials                                                   | AxonFramework-4.12.1/docs/_reference/modules/modelin             |
| 24 | AxonFramework-4.12.1/docs/_reference/modules/spring/pages                                                        | AxonFramework-4.12.1/docs/_reference/modules/sprin               |
| 25 | AxonFramework-4.12.1/docs/_reference/modules/spring/partials                                                     | AxonFramework-4.12.1/docs/_reference/modules/sprin               |
| 26 | AxonFramework-4.12.1/docs/_reference/modules/springboot-starter/pages                                            | AxonFramework-4.12.1/docs/_reference/modules/springboot-starte   |
| 27 | AxonFramework-4.12.1/docs/_reference/modules/springboot-starter/partials                                         | AxonFramework-4.12.1/docs/_reference/modules/springboot-starte   |
| 28 | AxonFramework-4.12.1/docs/_reference/modules/test/pages                                                          | AxonFramework-4.12.1/docs/_reference/modules/tes                 |
| 29 | AxonFramework-4.12.1/docs/_reference/modules/test/partials                                                       | AxonFramework-4.12.1/docs/_reference/modules/tes                 |

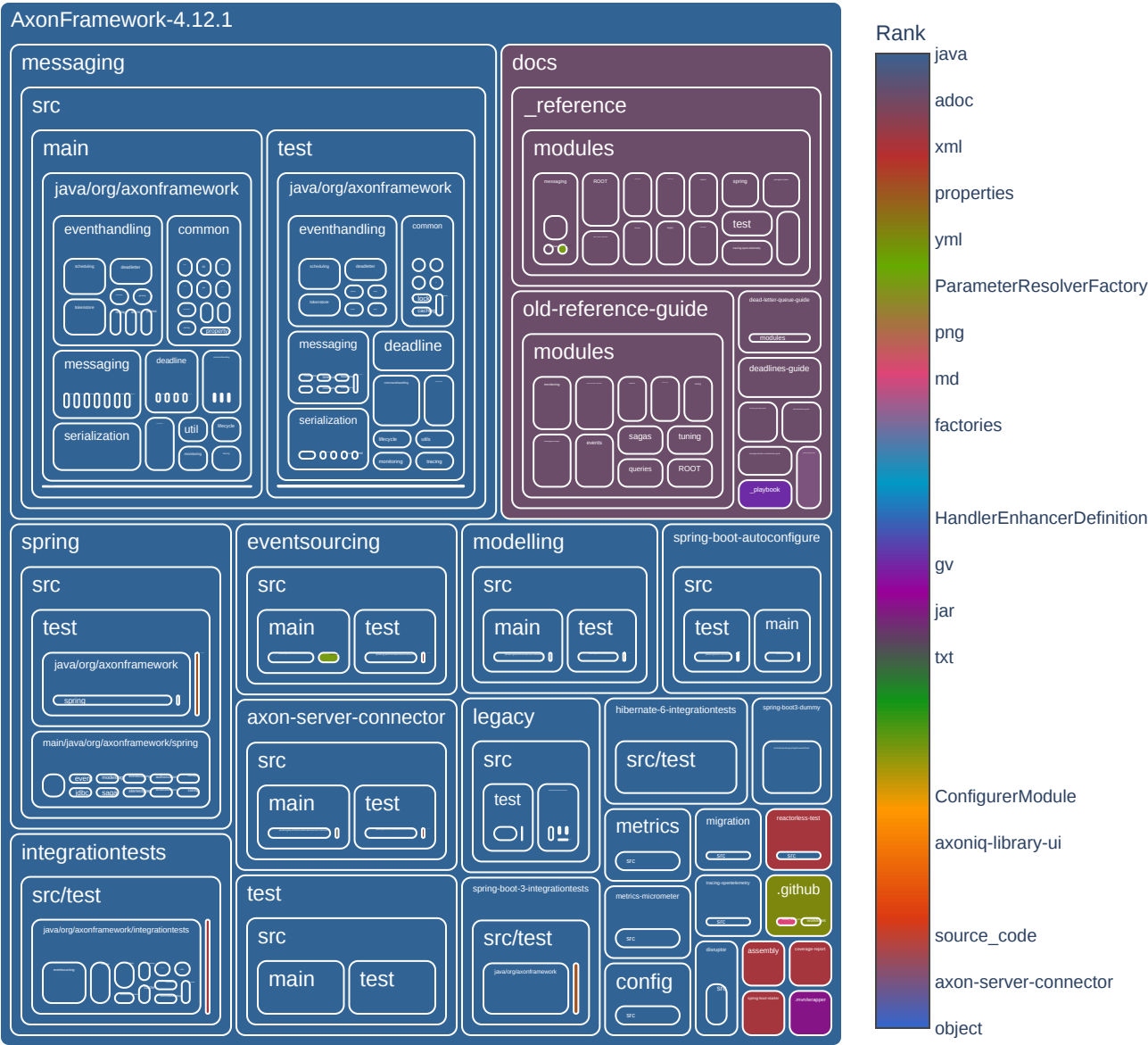
## Number of files per directory

Directories and their file count



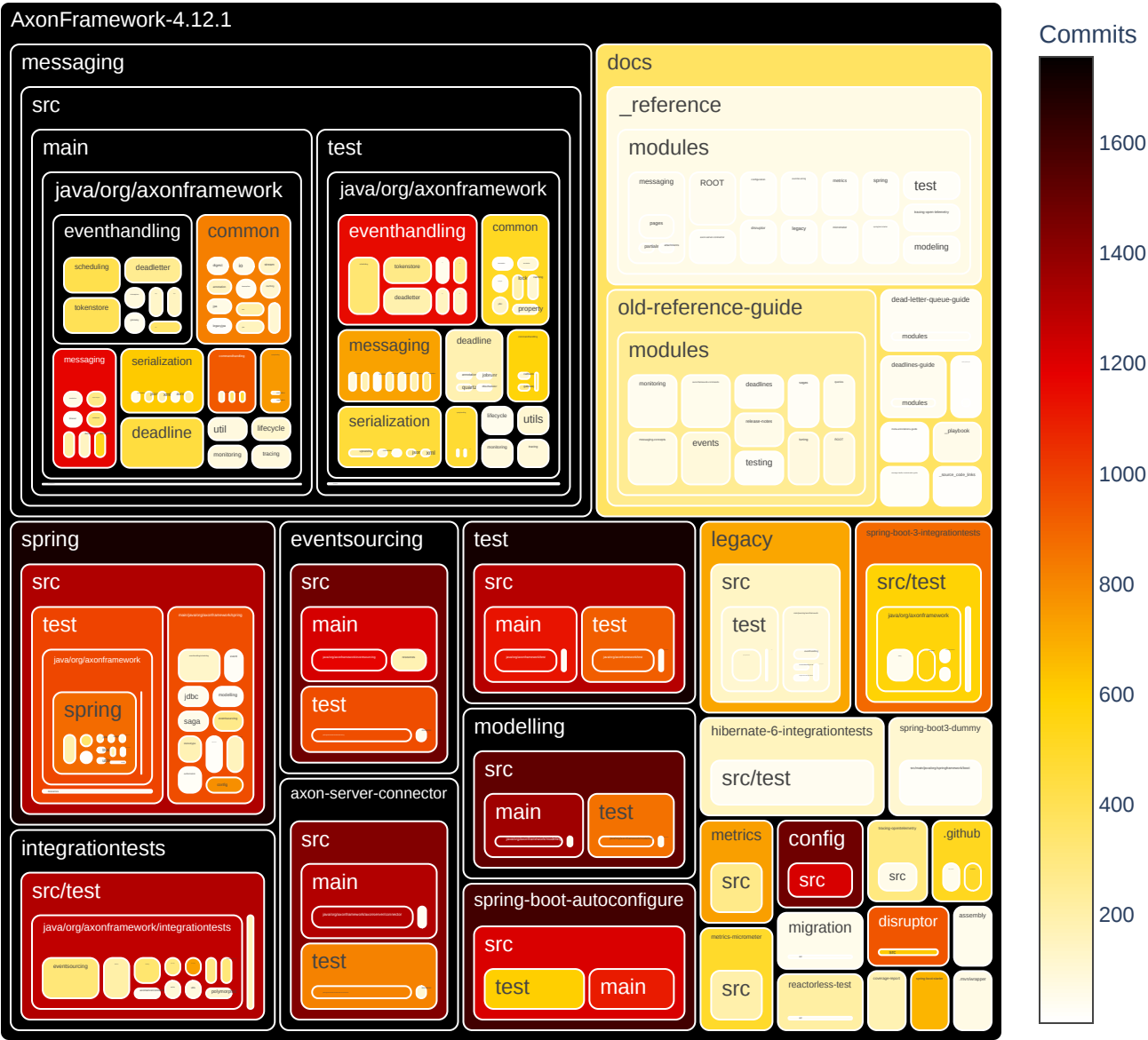
Most frequent file extension per directory

Most frequent file extension per directory



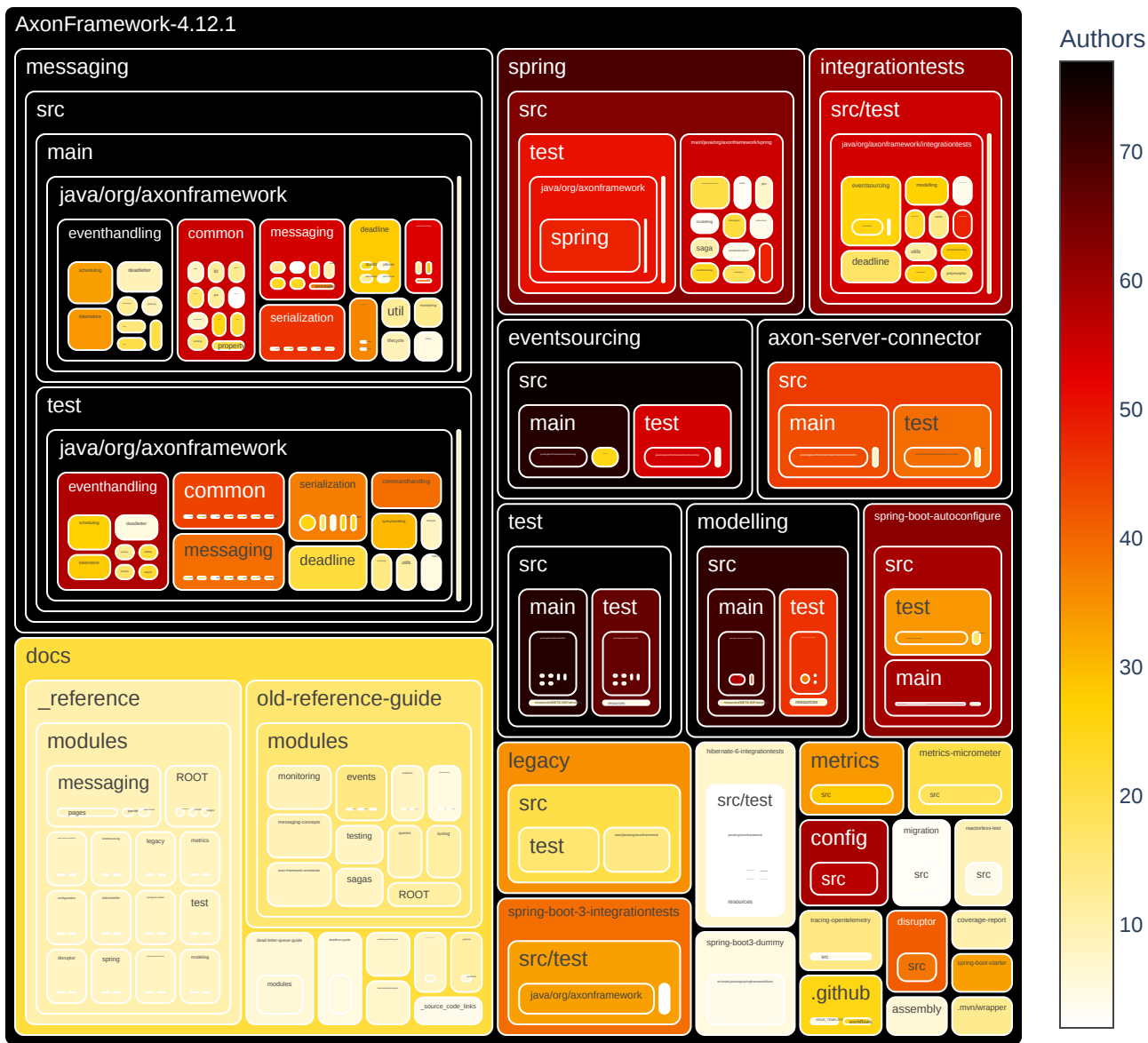
Number of commits per directory

Number of git commits



Number of distinct authors per directory

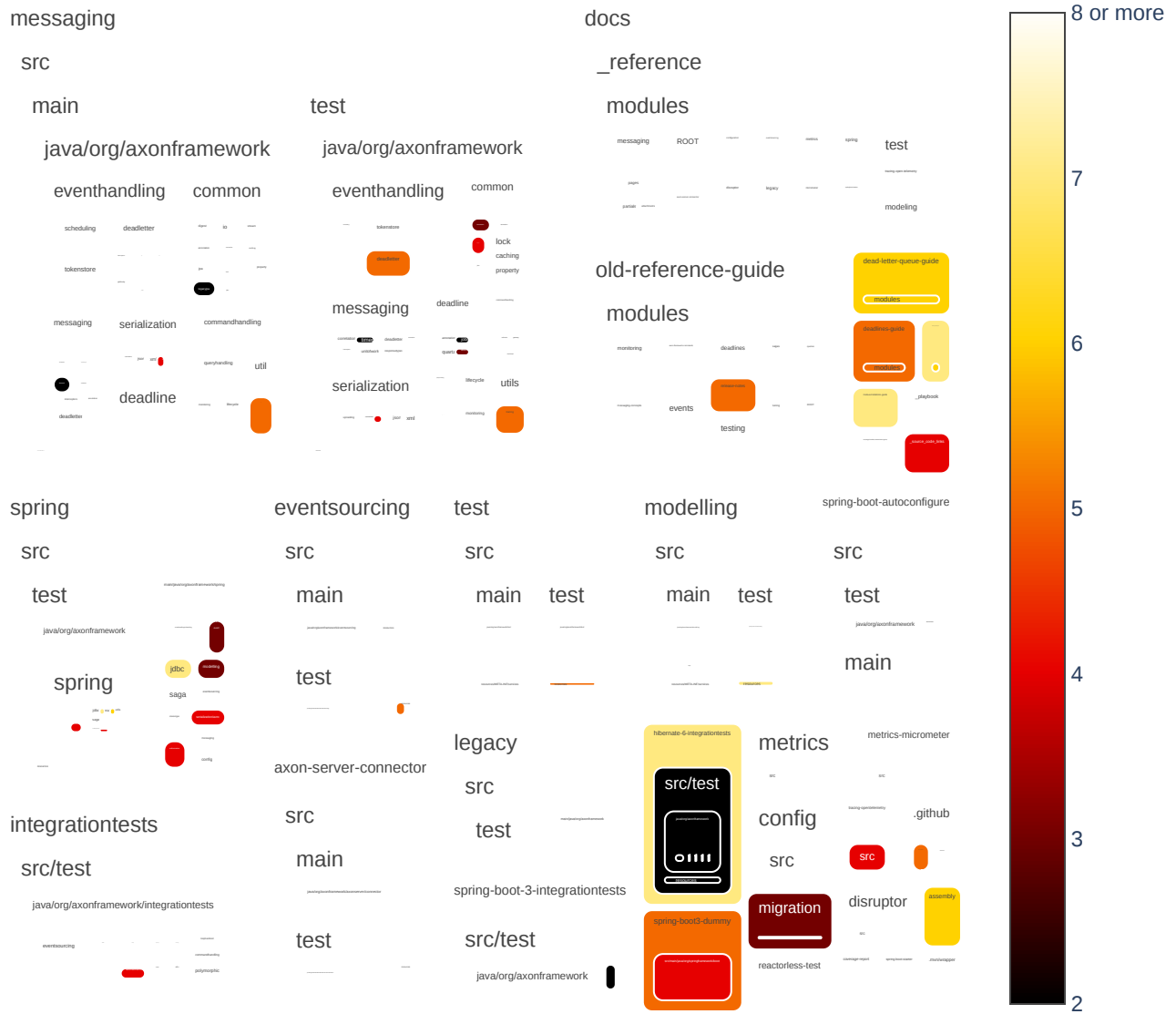
## Number of distinct commit authors



Directories with very few different authors

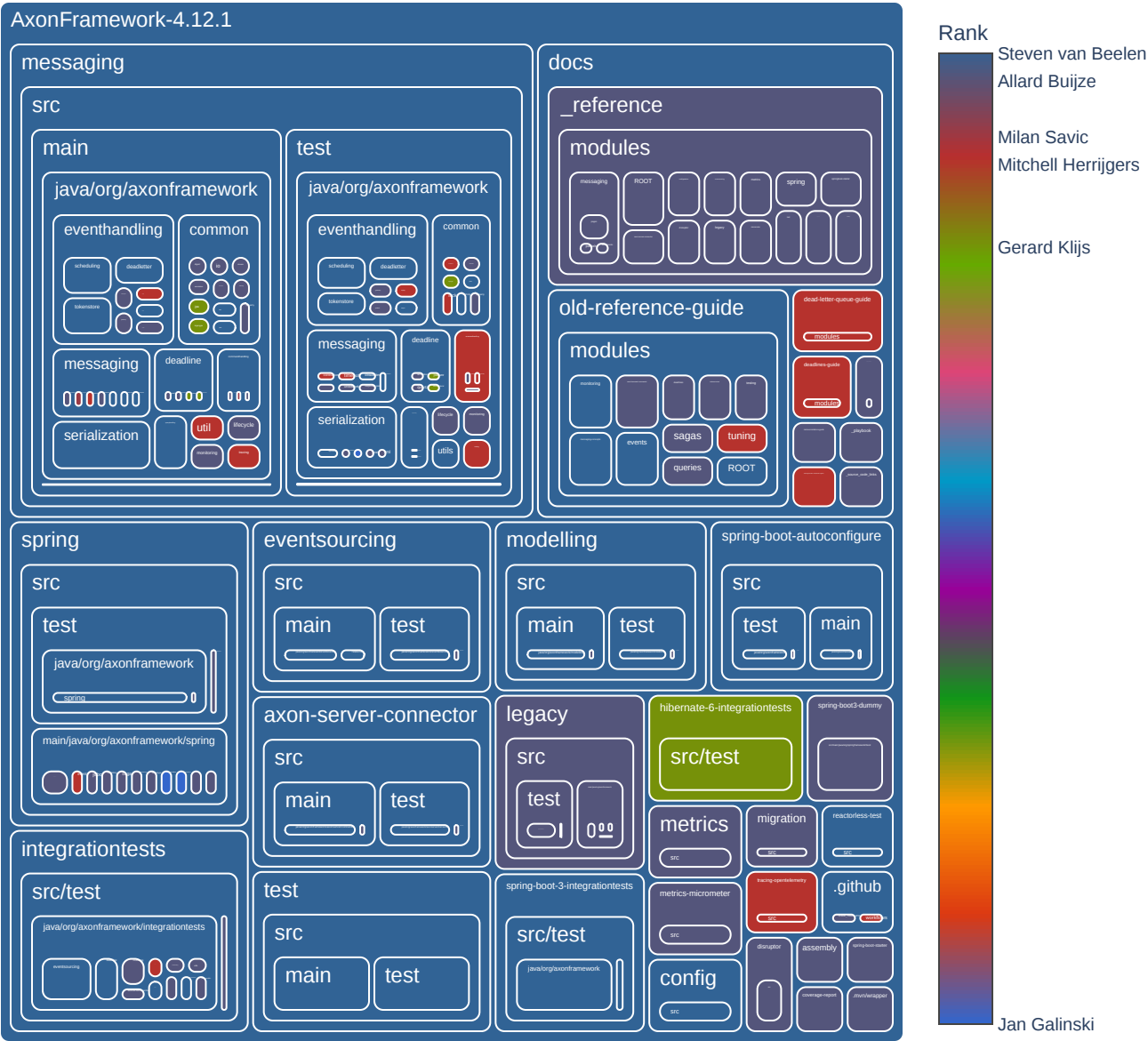
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.12.1



## Main author per directory

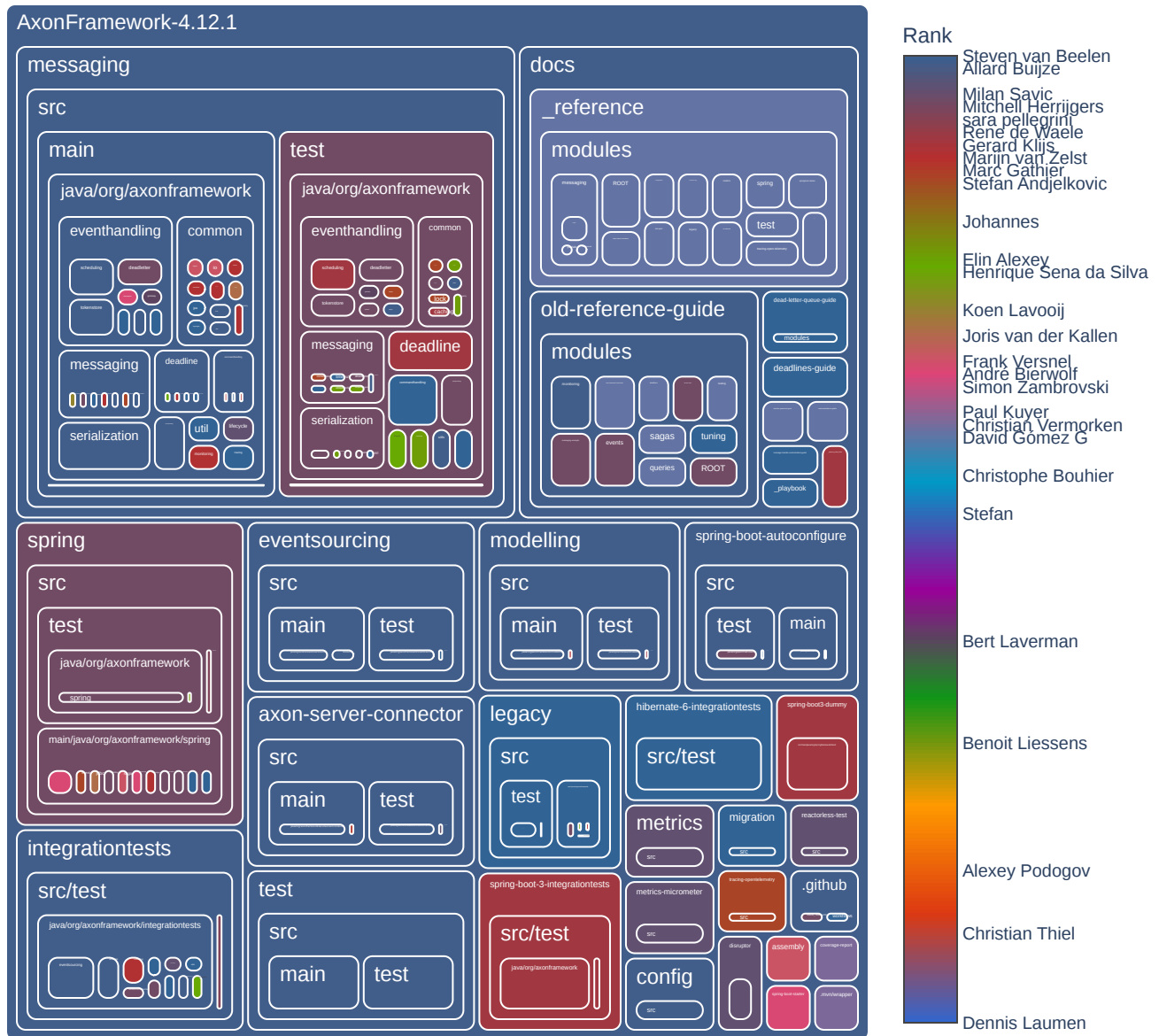
Main authors with highest number of commits



Second author per directory



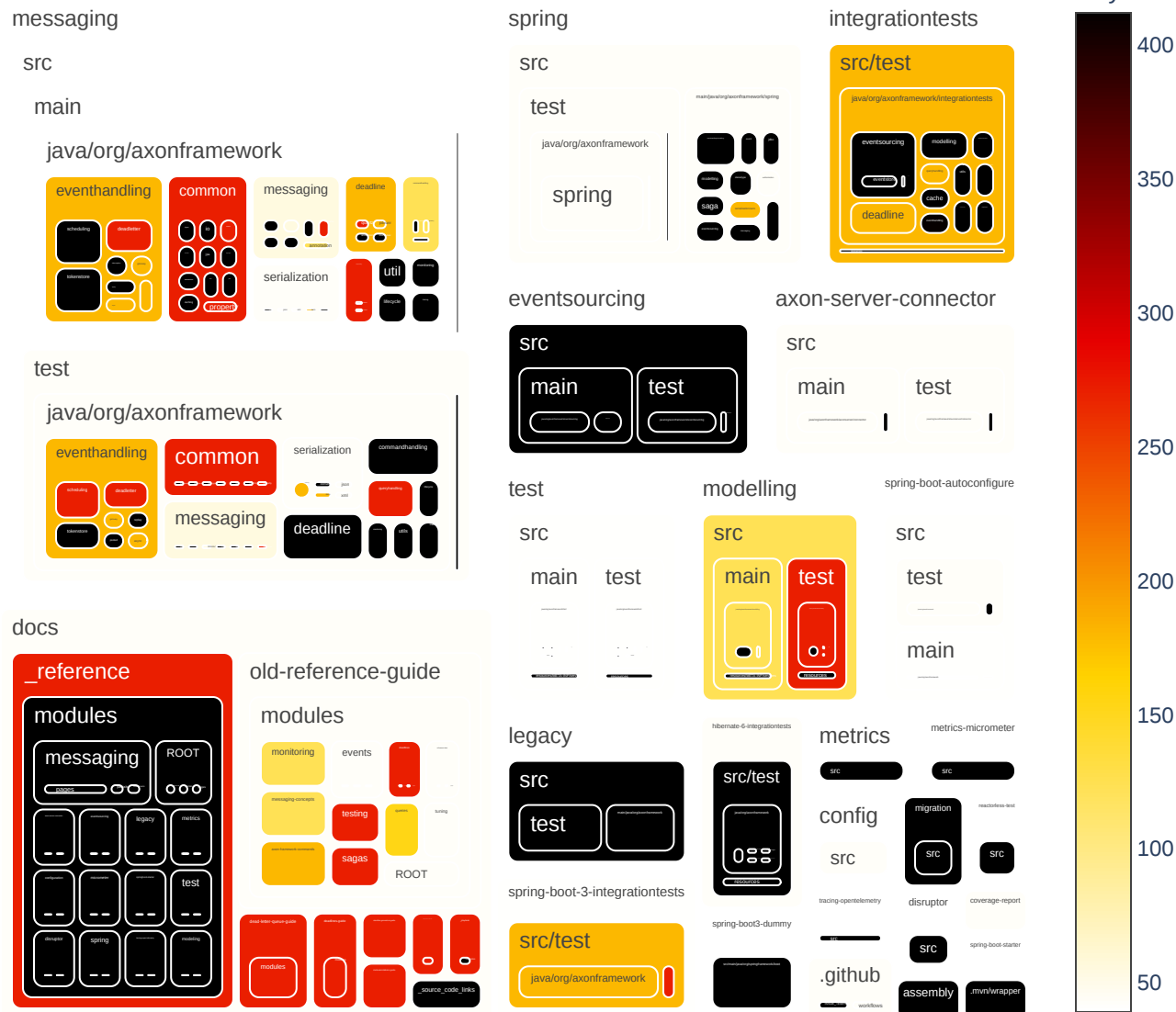
### Second author with the second highest number of commits



## Days since last commit per directory

Days since last commit

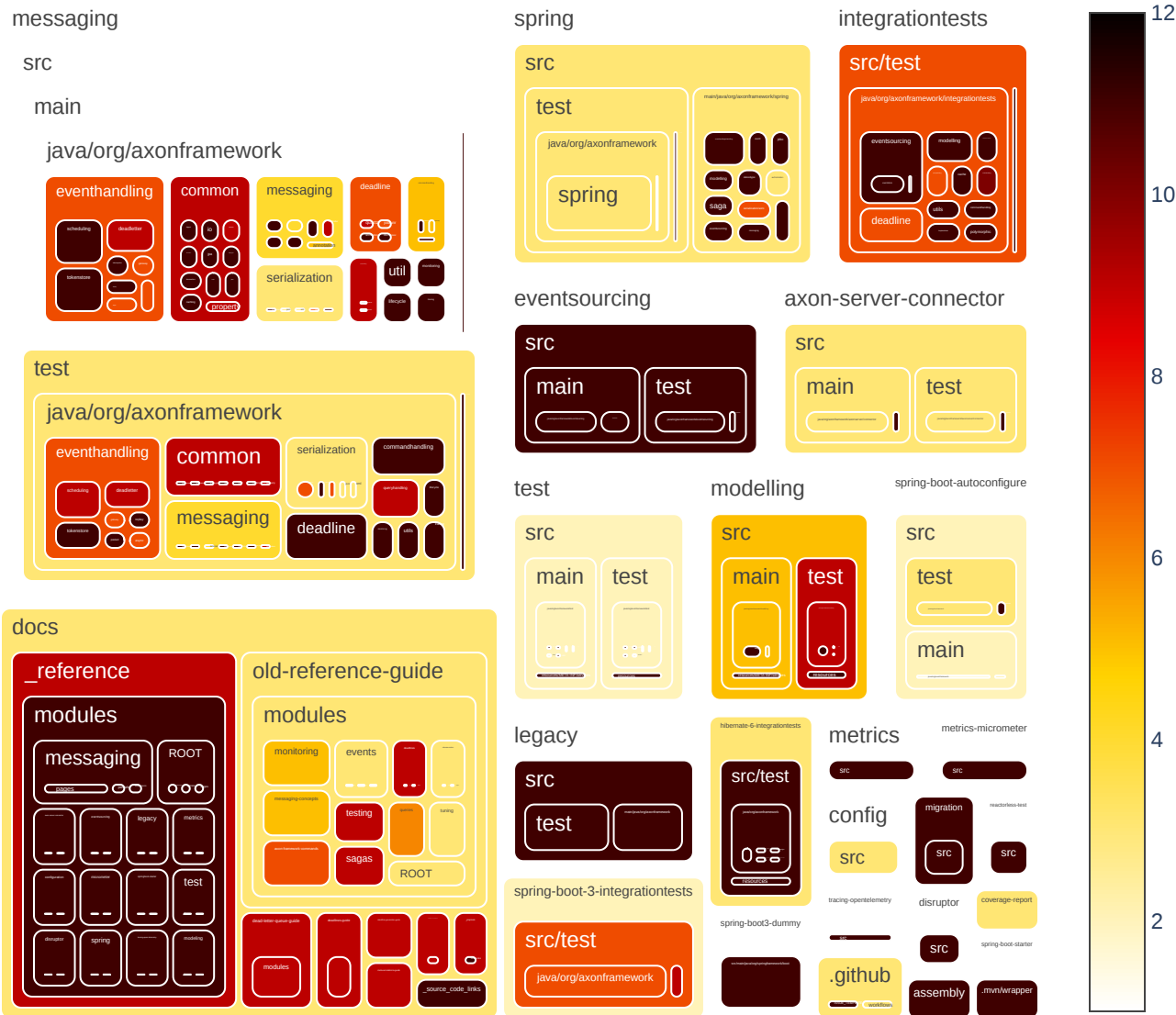
AxonFramework-4.12.1



Days since last commit per directory (ranked)

Rank of days since last commit

AxonFramework-4.12.1



Days since last file creation per directory

Days since last file creation

AxonFramework-4.12.1

messaging

src

main

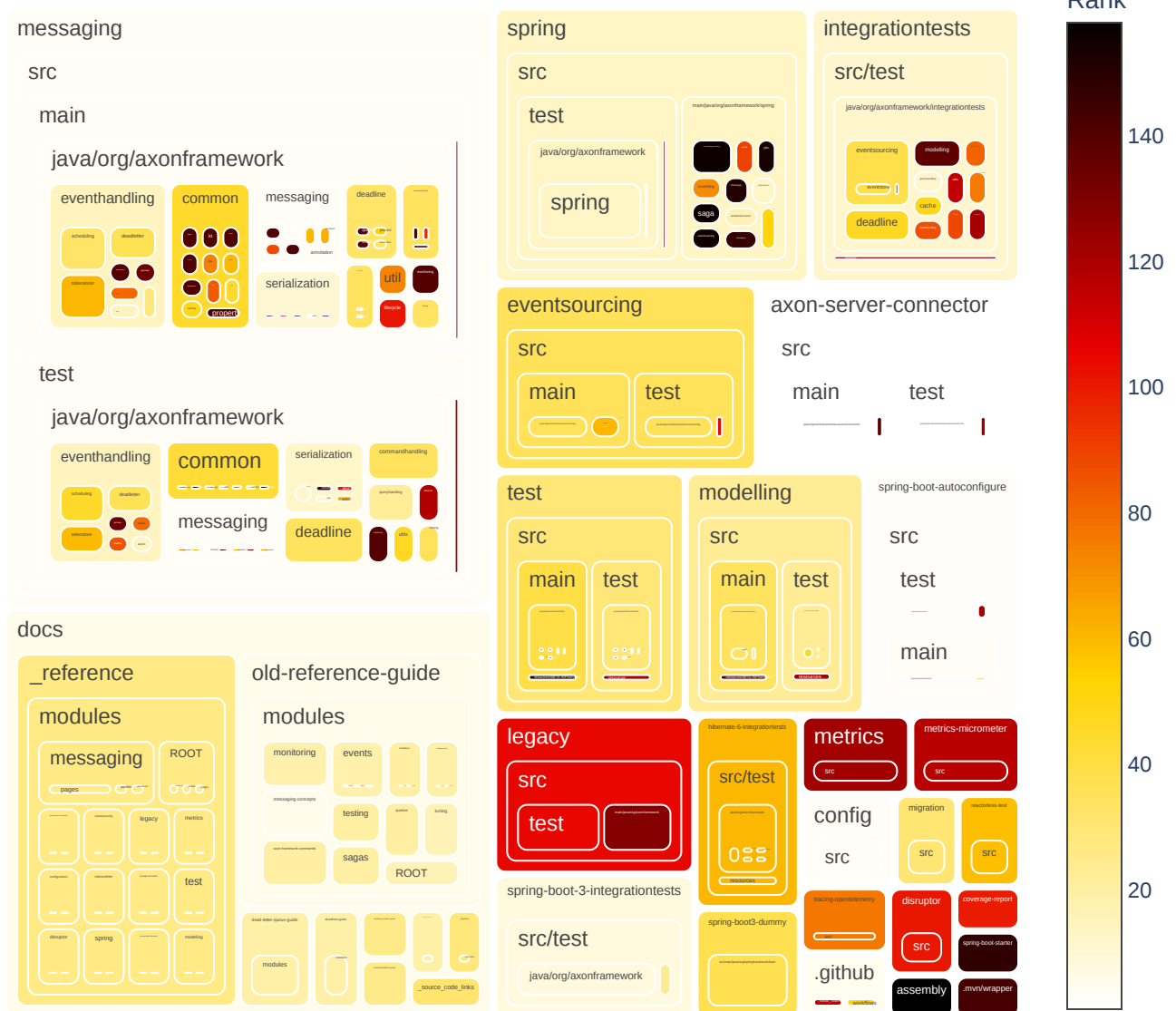
test



Days since last file creation per directory (ranked)

## Rank of days since last file creation

AxonFramework-4.12.1



## Days since last file modification per directory

Days since last file modification

AxonFramework-4.12.1



Days since last file modification per directory (ranked)

## Rank of days since last file modification

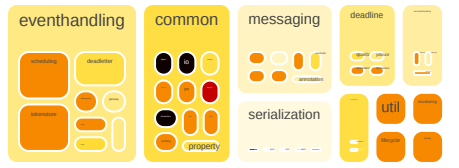
AxonFramework-4.12.1

messaging

src

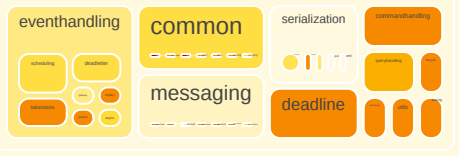
main

java/org/axonframework



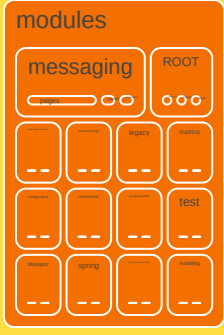
test

java/org/axonframework



docs

\_reference



old-reference-guide



spring

src

test

java/org/axonframework

spring

eventsourcing

src

main

test

test

src

main

test

legacy

src

test

spring-boot-3-integrationtests

src/test

java/org/axonframework

integrationtests

src/test

java/org/axonframework/integrationtests

deadline

axon-server-connector

src

main

test

modelling

src

main

test

spring-boot-autoconfigure

src

main

test

hibernate-6-integrationtests

src/test

088

spring-boot3-dummy

src

test

metrics

src

test

config

src

test

.github

assembly

min/wrapper

Rank

50

45

40

35

30

25

20

15

10

5

## Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

## Preview data

Sum of commits that changed more than 30 files (each) = 973

Max changed files with one commit = 2178

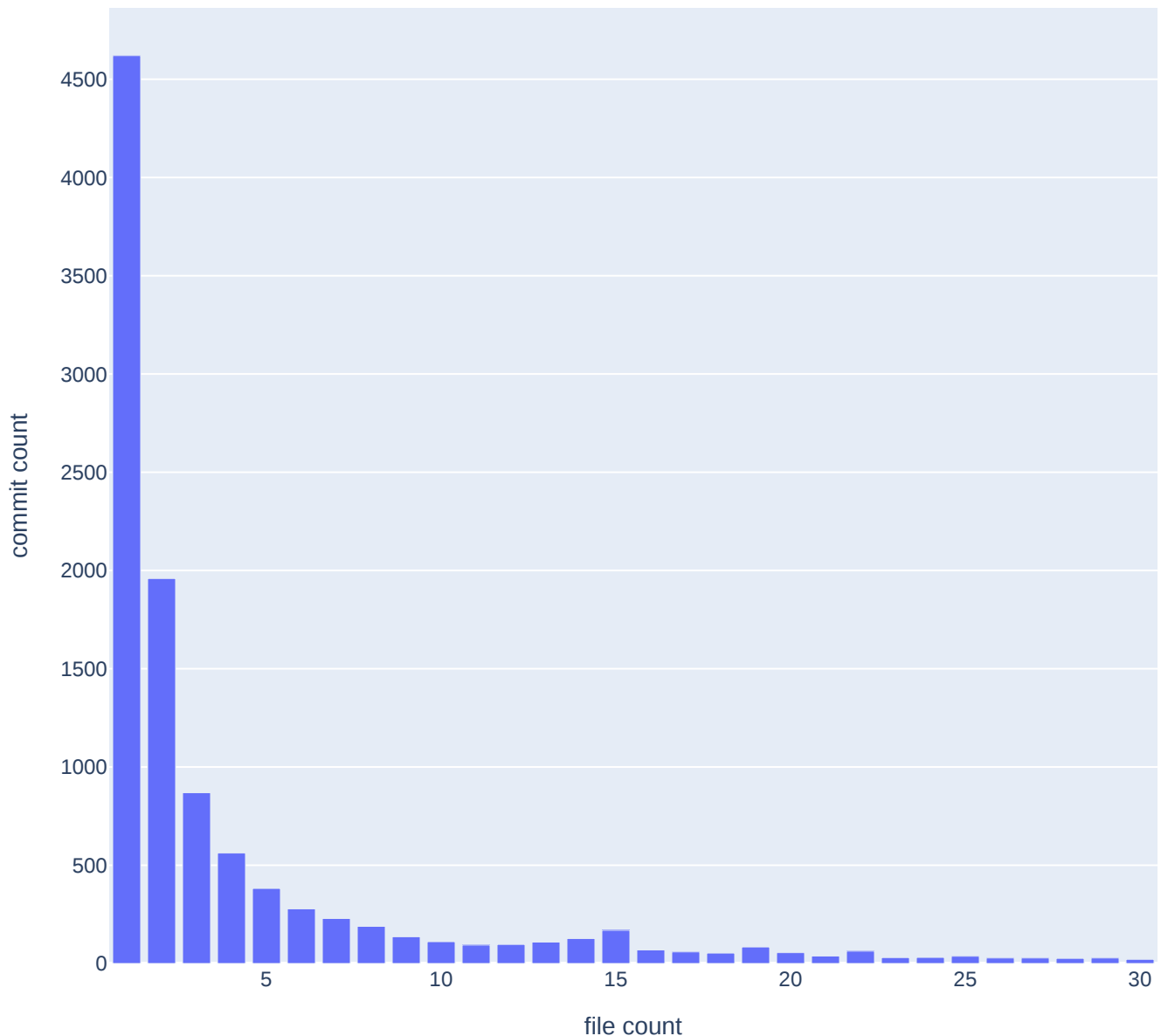
|       | filesPerCommit | commitCount |
|-------|----------------|-------------|
| count | 334.000000     | 334.000000  |
| mean  | 343.449102     | 34.592814   |
| std   | 439.862566     | 281.096056  |
| min   | 1.000000       | 1.000000    |
| 25%   | 84.250000      | 1.000000    |
| 50%   | 185.000000     | 2.000000    |
| 75%   | 426.500000     | 5.000000    |
| max   | 2178.000000    | 4620.000000 |

|    | filesPerCommit | commitCount |
|----|----------------|-------------|
| 0  | 1              | 4620        |
| 1  | 2              | 1959        |
| 2  | 3              | 869         |
| 3  | 4              | 562         |
| 4  | 5              | 382         |
| 5  | 6              | 278         |
| 6  | 7              | 229         |
| 7  | 8              | 189         |
| 8  | 9              | 136         |
| 9  | 10             | 111         |
| 10 | 11             | 94          |
| 11 | 12             | 97          |
| 12 | 13             | 109         |
| 13 | 14             | 127         |
| 14 | 15             | 169         |
| 15 | 16             | 69          |
| 16 | 17             | 59          |
| 17 | 18             | 53          |
| 18 | 19             | 84          |
| 19 | 20             | 55          |
| 20 | 21             | 38          |
| 21 | 22             | 63          |
| 22 | 23             | 30          |
| 23 | 24             | 31          |
| 24 | 25             | 37          |
| 25 | 26             | 28          |
| 26 | 27             | 28          |
| 27 | 28             | 26          |
| 28 | 29             | 28          |
| 29 | 30             | 21          |

Bar chart with the number of files per commit distribution



## Changed files per commit



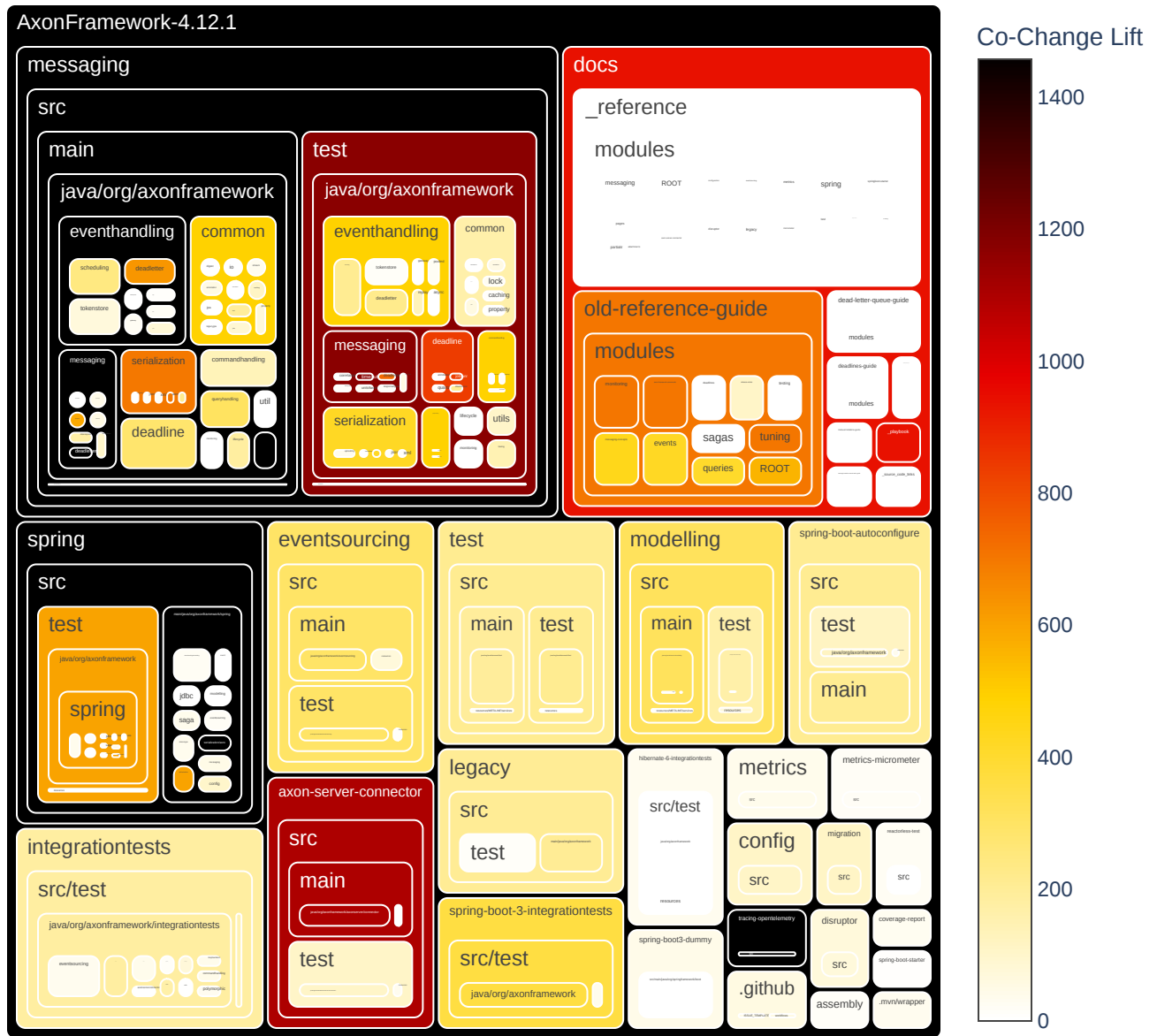
## Pairwise Changed Files

This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

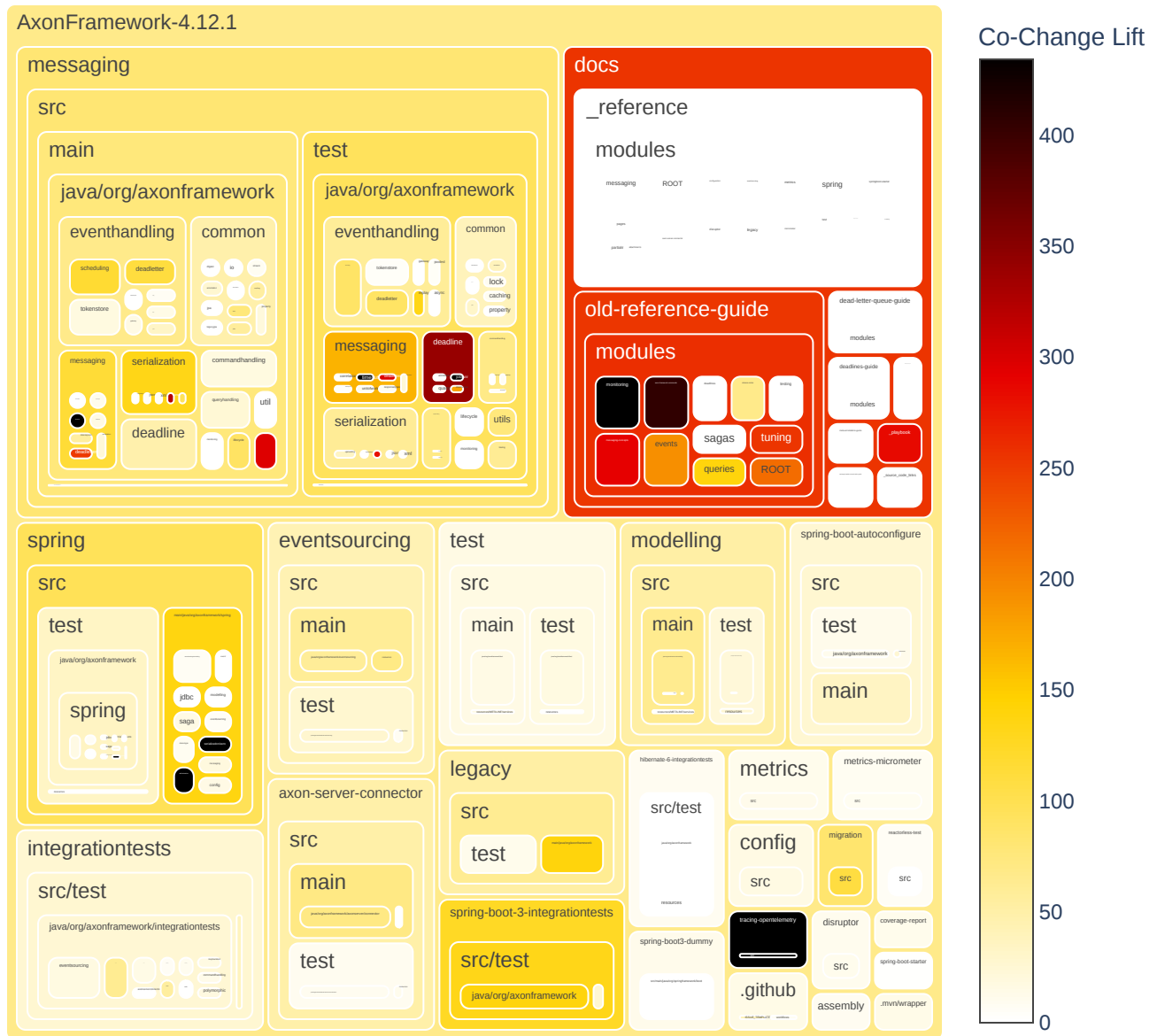
- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

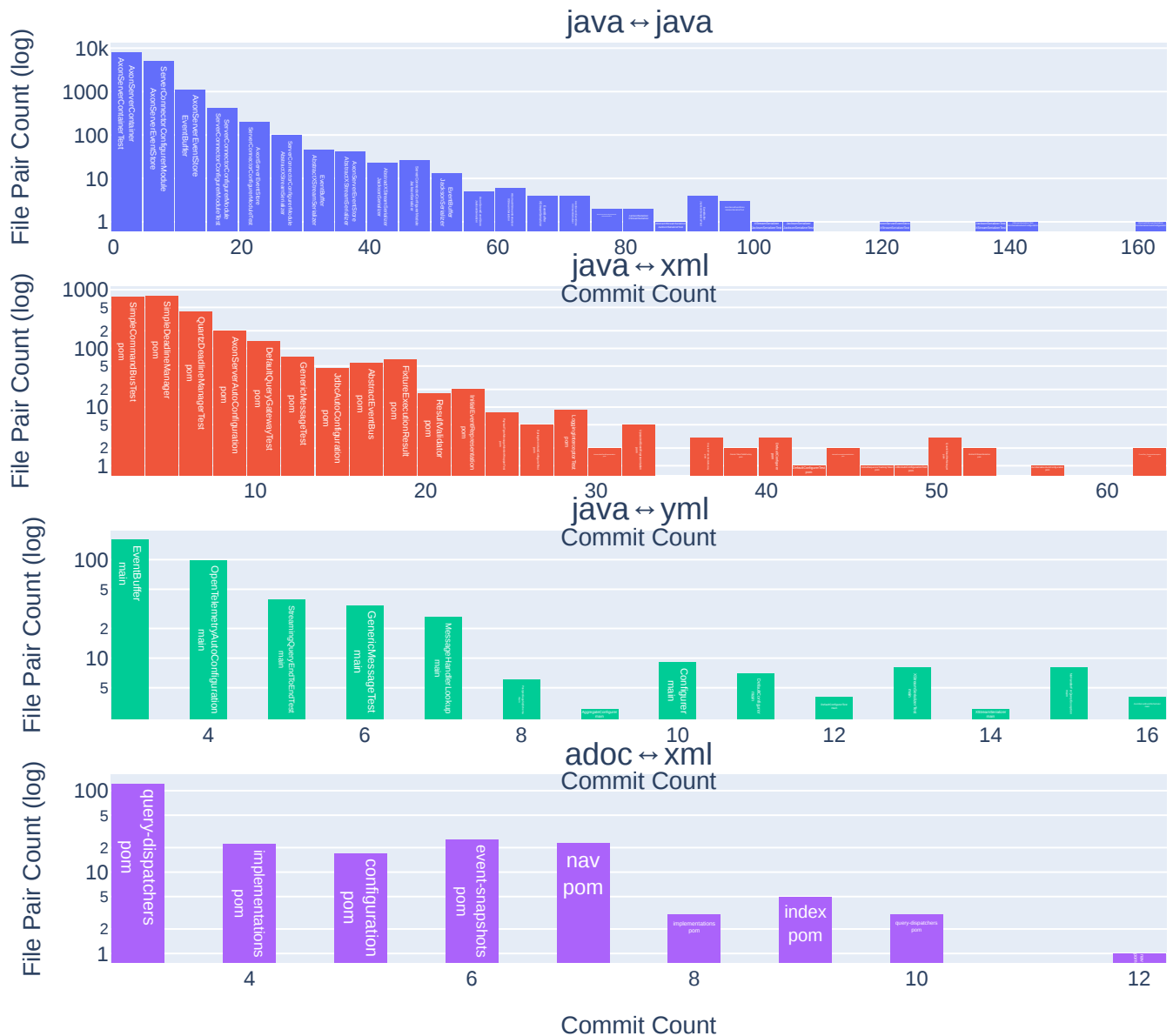


|   | fileExtensionPair | fileExtensionPairCount |
|---|-------------------|------------------------|
| 0 | java↔java         | 15189                  |
| 1 | java↔xml          | 2637                   |
| 2 | java↔yml          | 408                    |
| 3 | adoc↔xml          | 221                    |

## Files changed together by commit count

|    | fileExtensionPair | updateCommitCount | GroupRank | filePair                                                                                     |
|----|-------------------|-------------------|-----------|----------------------------------------------------------------------------------------------|
| 0  | java↔java         | 163               | 1         | AxonServerQueryBus↔AxonServerCommandBus                                                      |
| 1  | java↔java         | 144               | 2         | AxonServerQueryBus↔AxonServerQueryBusTest                                                    |
| 2  | java↔java         | 135               | 3         | SpringAxonAutoConfigurerTest↔SpringAxonAutoConfigurer spring/src/tes                         |
| 3  | java↔java         | 121               | 4         | TrackingEventProcessorTest↔TrackingEventProcessor integrationtests/src/test/java/org/axonfra |
| 4  | java↔java         | 107               | 5         | SpringAxonAutoConfigurer↔AxonConfiguration spi                                               |
| 5  | java↔java         | 100               | 6         | EventProcessingModule↔EventProcessingModuleTest                                              |
| 6  | java↔java         | 96                | 7         | SagaTestFixture↔AggregateTestFixture                                                         |
| 7  | java↔java         | 95                | 8         | AxonServerQueryBusTest↔AxonServerCommandBusTest                                              |
| 8  | java↔java         | 94                | 9         | AggregateTestFixture↔FixtureConfiguration                                                    |
| 9  | java↔java         | 92                | 10        | AggregateTestFixture↔SpringAxonAutoConfigurer test/                                          |
| 10 | java↔xml          | 62                | 1         | AggregateTestFixture↔pom                                                                     |
| 11 | java↔xml          | 57                | 2         | AggregateTestFixture↔pom                                                                     |
| 12 | java↔xml          | 53                | 3         | AggregateTestFixture↔pom                                                                     |
| 13 | java↔xml          | 51                | 4         | SpringAxonAutoConfigurer↔pom                                                                 |
| 14 | java↔xml          | 48                | 5         | SagaTestFixture↔pom                                                                          |
| 15 | java↔xml          | 46                | 6         | SpringAxonAutoConfigurer↔pom                                                                 |
| 16 | java↔xml          | 45                | 7         | SpringAxonAutoConfigurerTest↔pom                                                             |
| 17 | java↔xml          | 44                | 8         | SpringAxonAutoConfigurerTest↔pom                                                             |
| 18 | java↔xml          | 42                | 9         | AggregateTestFixture↔pom                                                                     |
| 19 | java↔xml          | 41                | 10        | AxonConfiguration↔pom                                                                        |
| 20 | java↔yaml         | 16                | 1         | JobRunrDeadlineManagerTest↔main                                                              |
| 21 | java↔yaml         | 15                | 2         | JobRunrEventSchedulerTest↔main                                                               |
| 22 | java↔yaml         | 14                | 3         | TrackingEventProcessor↔main                                                                  |
| 23 | java↔yaml         | 13                | 4         | SimpleQueryBus↔main                                                                          |
| 24 | java↔yaml         | 12                | 5         | SimpleEventHandlerInvokerTest↔main                                                           |
| 25 | java↔yaml         | 11                | 6         | AxonServerEventStore↔main                                                                    |
| 26 | java↔yaml         | 10                | 7         | PooledStreamingEventProcessor↔main                                                           |
| 27 | java↔yaml         | 9                 | 8         | AxonServerQueryBusTest↔main                                                                  |
| 28 | java↔yaml         | 8                 | 9         | DefaultConfigurer↔main                                                                       |
| 29 | java↔yaml         | 7                 | 10        | AggregateAnnotationCommandHandler↔main                                                       |
| 30 | adoc↔xml          | 12                | 1         | minor-releases↔pom                                                                           |
| 31 | adoc↔xml          | 10                | 2         | rdbms-tuning↔pom                                                                             |
| 32 | adoc↔xml          | 9                 | 3         | minor-releases↔pom                                                                           |
| 33 | adoc↔xml          | 8                 | 4         | minor-releases↔pom                                                                           |
| 34 | adoc↔xml          | 7                 | 5         | minor-releases↔pom                                                                           |
| 35 | adoc↔xml          | 6                 | 6         | nav↔pom                                                                                      |
| 36 | adoc↔xml          | 5                 | 7         | event-versioning↔pom                                                                         |
| 37 | adoc↔xml          | 4                 | 8         | rdbms-tuning↔pom                                                                             |
| 38 | adoc↔xml          | 3                 | 9         | query-dispatchers↔pom                                                                        |

## Co-Changed Files by their commit count

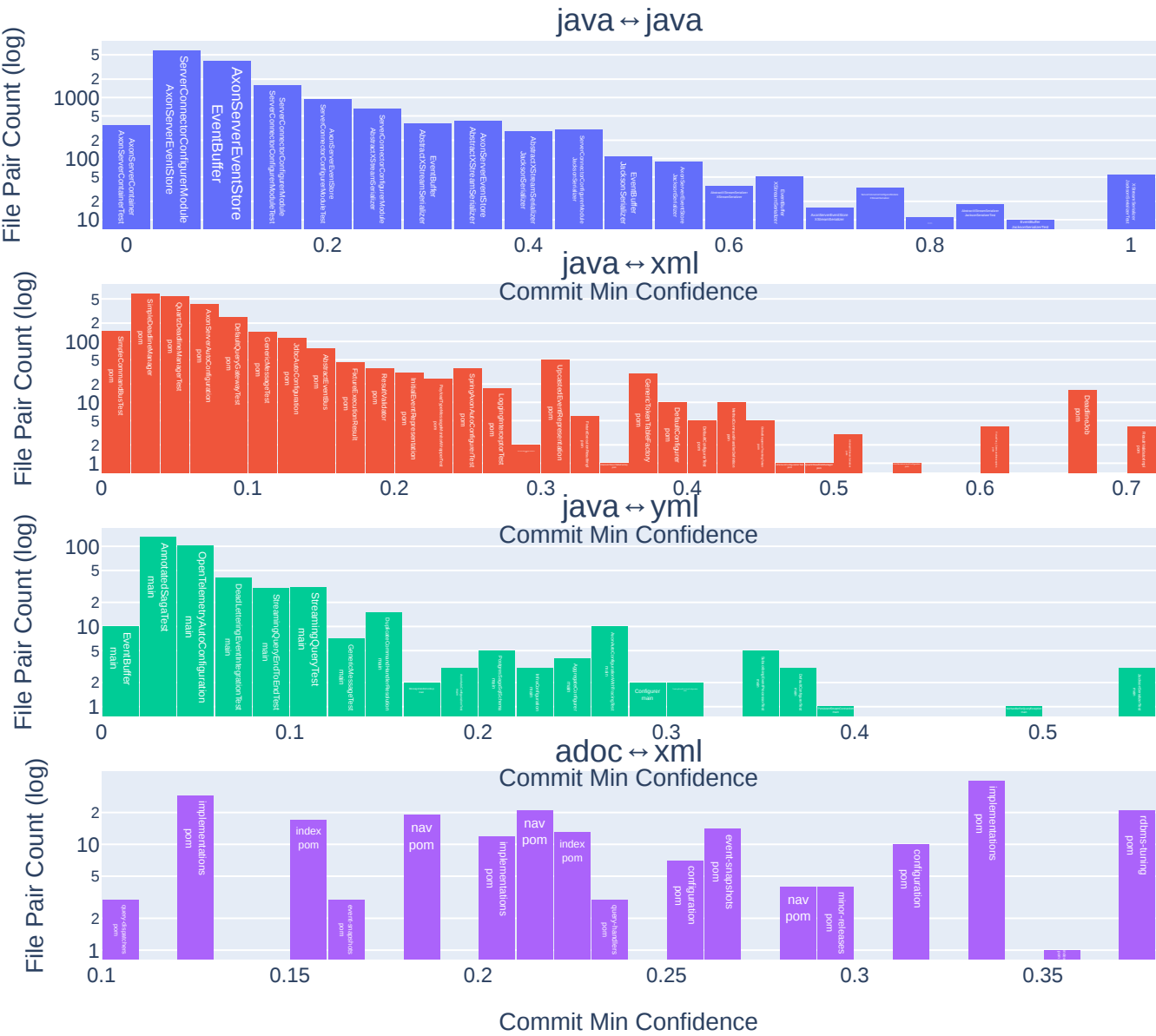


## Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

|    | fileExtensionPair | updateCommitMinConfidence | GroupRank | filePair                                                               |
|----|-------------------|---------------------------|-----------|------------------------------------------------------------------------|
| 0  | java↔java         | 1.000000                  | 1         | AxonTimeLimitedTask↔UnitOfWorkTimeoutInterceptor                       |
| 1  | java↔java         | 0.909091                  | 2         | AvroUtil↔SpecificRecordBaseSerializerStrategy                          |
| 2  | java↔java         | 0.900000                  | 3         | AxonTimeLimitedTask↔TimeoutWrappedMessageHandlingMember                |
| 3  | java↔java         | 0.875000                  | 4         | Decisions↔DoNotEnqueue                                                 |
| 4  | java↔java         | 0.857143                  | 5         | AvroSerializer↔AvroSerializerStrategy                                  |
| 5  | java↔java         | 0.833333                  | 6         | AvroSerializerStrategy↔ByteArrayToGenericRecordConverter               |
| 6  | java↔java         | 0.800000                  | 7         | DeadLetteringEventIntegrationTest↔InMemoryDeadLetteringIntegrationTest |
| 7  | java↔java         | 0.785714                  | 8         | AvroSerializer↔AvroSerializerTest                                      |
| 8  | java↔java         | 0.777778                  | 9         | JobrunrDeadlineManagerTest↔JobRunrEventSchedulerTest                   |
| 9  | java↔java         | 0.750000                  | 10        | AxonServerAutoConfiguration↔PersistentStreamMessageSource              |
| 10 | java↔xml          | 0.703704                  | 1         | JobRunrEventSchedulerTest↔pom                                          |
| 11 | java↔xml          | 0.666667                  | 2         | JobRunrEventSchedulerTest↔pom                                          |
| 12 | java↔xml          | 0.600000                  | 3         | ChainingConverter↔pom                                                  |
| 13 | java↔xml          | 0.545455                  | 4         | EventStoreTestUtils↔pom                                                |
| 14 | java↔xml          | 0.512821                  | 5         | DefaultEventMessageConverter↔pom                                       |
| 15 | java↔xml          | 0.500000                  | 6         | ChainingConverter↔pom                                                  |
| 16 | java↔xml          | 0.487179                  | 7         | DefaultEventMessageConverter↔pom                                       |
| 17 | java↔xml          | 0.461538                  | 8         | DefaultEventMessageConverter↔pom                                       |
| 18 | java↔xml          | 0.454545                  | 9         | JpaEventStoreAutoConfigurationWithoutAxonServerTest↔pom                |
| 19 | java↔xml          | 0.448276                  | 10        | JobrunrDeadlineManagerTest↔pom                                         |
| 20 | java↔yaml         | 0.555556                  | 1         | JobRunrEventSchedulerTest↔main                                         |
| 21 | java↔yaml         | 0.481481                  | 2         | JobRunrEventSchedulerTest↔antora                                       |
| 22 | java↔yaml         | 0.392857                  | 3         | AvroSerializerAutoConfiguration↔antora                                 |
| 23 | java↔yaml         | 0.375000                  | 4         | AxonTimeLimitedTaskTest↔main                                           |
| 24 | java↔yaml         | 0.357143                  | 5         | AvroSerializerAutoConfiguration↔main                                   |
| 25 | java↔yaml         | 0.340909                  | 6         | AbstractDeadlineManagerTestSuite↔main                                  |
| 26 | java↔yaml         | 0.300000                  | 7         | ChainingConverter↔main                                                 |
| 27 | java↔yaml         | 0.295455                  | 8         | AbstractDeadlineManagerTestSuite↔antora                                |
| 28 | java↔yaml         | 0.288889                  | 9         | JobrunrDeadlineManagerTest↔antora                                      |
| 29 | java↔yaml         | 0.277778                  | 10        | SpringAxonConfiguration↔main                                           |
| 30 | adoc↔xml          | 0.375000                  | 1         | index↔pom                                                              |
| 31 | adoc↔xml          | 0.352941                  | 2         | minor-releases↔pom                                                     |
| 32 | adoc↔xml          | 0.333333                  | 3         | nav↔pom                                                                |
| 33 | adoc↔xml          | 0.315789                  | 4         | query-dispatchers↔pom                                                  |
| 34 | adoc↔xml          | 0.312500                  | 5         | query-handlers↔pom                                                     |
| 35 | adoc↔xml          | 0.291667                  | 6         | event-snapshots↔pom                                                    |
| 36 | adoc↔xml          | 0.285714                  | 7         | implementations↔pom                                                    |
| 37 | adoc↔xml          | 0.281250                  | 8         | rbms-tuning↔pom                                                        |
| 38 | adoc↔xml          | 0.264706                  | 9         | minor-releases↔pom                                                     |
| 39 | adoc↔xml          | 0.263158                  | 10        | event-versioning↔pom                                                   |

Co-Changed Files by their commit min confidence

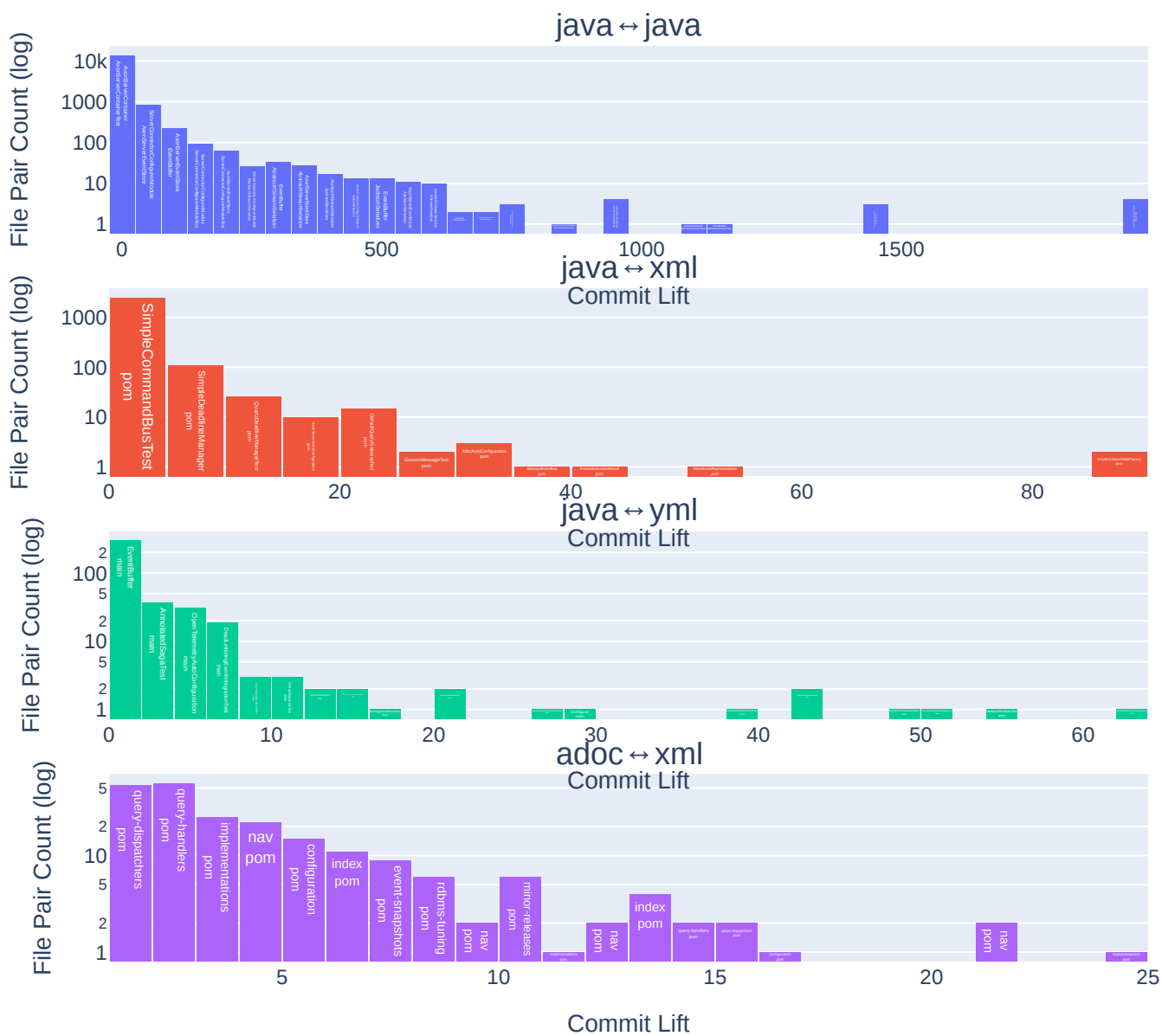


Files changed together by commit lift



| file | ExtensionPair | updateCommitLift | GroupRank |                                                                                           |
|------|---------------|------------------|-----------|-------------------------------------------------------------------------------------------|
| 0    | java↔java     | 1942.666667      | 1         | AvroSchemaPackages↔Av                                                                     |
| 1    | java↔java     | 1457.000000      | 2         | DefaultEventProcessorSpanFactory↔DefaultEventProcessorS                                   |
| 2    | java↔java     | 1165.600000      | 3         | TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeoul                                 |
| 3    | java↔java     | 1092.750000      | 4         | AxonServerEventStoreFactory↔AxonServerEventS                                              |
| 4    | java↔java     | 971.333333       | 5         | ClasspathAvroSchemaLoader↔SpecificRecordBaseClasspathAvrc                                 |
| 5    | java↔java     | 832.571429       | 6         | DeadlineDetailsSerializationTest↔JobRunnrDeadlineMan                                      |
| 6    | java↔java     | 728.500000       | 7         | EnqueuePolicy↔NoSuchDeadl                                                                 |
| 7    | java↔java     | 699.360000       | 8         | AxonServerNonTransientRemoteCommandHandlingException↔AxonServerNonTransientRemoteQueryHar |
| 8    | java↔java     | 693.809524       | 9         | AvroSerializerStrategy↔ByteArrayToGenericR                                                |
| 9    | java↔java     | 647.555556       | 10        | GenericDeadLetter                                                                         |
| 10   | java↔xml      | 86.726190        | 1         | ContextAddingInter                                                                        |
| 11   | java↔xml      | 50.590278        | 2         | UpstreamAwareStreamOb:                                                                    |
| 12   | java↔xml      | 42.695971        | 3         | JobRunnrEventSche                                                                         |
| 13   | java↔xml      | 35.321212        | 4         | EventStoreTestUt                                                                          |
| 14   | java↔xml      | 33.996667        | 5         | TestAggregateRoot↔benchmark-                                                              |
| 15   | java↔xml      | 33.167480        | 6         | Message↔benchmark-                                                                        |
| 16   | java↔xml      | 30.118863        | 7         | JobRunnrEventSche                                                                         |
| 17   | java↔xml      | 27.752381        | 8         | MessagingCommandHandler↔benchmark-                                                        |
| 18   | java↔xml      | 26.076063        | 9         | JobRunnrEventSche                                                                         |
| 19   | java↔xml      | 24.283333        | 10        | AbstractResourceInject                                                                    |
| 20   | java↔yaml     | 62.357202        | 1         | JobRunnrEventSchedt                                                                       |
| 21   | java↔yaml     | 54.637500        | 2         | AbstractResourceInjector↔                                                                 |
| 22   | java↔yaml     | 50.879365        | 3         | AvroSerializerAutoConfig                                                                  |
| 23   | java↔yaml     | 48.566667        | 4         | AxonTimeLimitedT                                                                          |
| 24   | java↔yaml     | 43.753754        | 5         | JobRunnrEventSchedulerTest↔slack-rele                                                     |
| 25   | java↔yaml     | 43.710000        | 6         | ConfigurationScopeAwareProvider↔                                                          |
| 26   | java↔yaml     | 38.264646        | 7         | AbstractDeadlineManagerTe                                                                 |
| 27   | java↔yaml     | 29.028352        | 8         | JobrunnrDeadlineMana                                                                      |
| 28   | java↔yaml     | 26.848894        | 9         | AbstractDeadlineManagerTestSuite↔slack-rele                                               |
| 29   | java↔yaml     | 21.479115        | 10        | JpaEventStoreAutoConfigurationWithoutAxonServerTest↔slack-rele                            |
| 30   | adoc↔xml      | 24.016484        | 1         |                                                                                           |
| 31   | adoc↔xml      | 21.347985        | 2         |                                                                                           |
| 32   | adoc↔xml      | 16.941860        | 3         |                                                                                           |
| 33   | adoc↔xml      | 15.059432        | 4         | conf                                                                                      |
| 34   | adoc↔xml      | 14.667785        | 5         |                                                                                           |
| 35   | adoc↔xml      | 14.231990        | 6         |                                                                                           |
| 36   | adoc↔xml      | 13.723705        | 7         | implem                                                                                    |
| 37   | adoc↔xml      | 13.185520        | 8         | minor                                                                                     |
| 38   | adoc↔xml      | 13.038031        | 9         | conf                                                                                      |
| 39   | adoc↔xml      | 12.008242        | 10        | query                                                                                     |

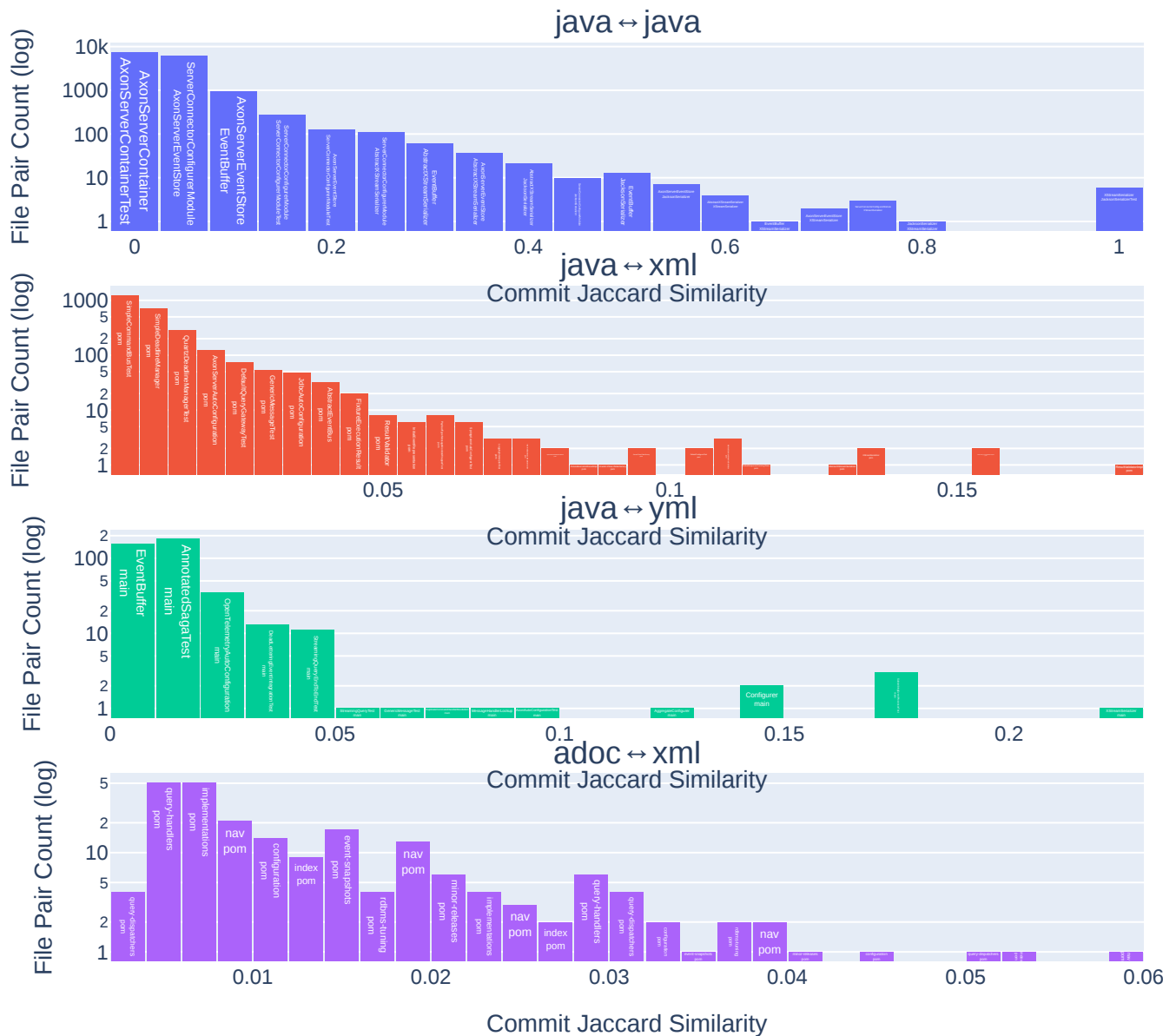
Co-Changed Files by their commit lift



Files changed together by commit Jaccard similarity

| fileExtensionPair | updateCommitJaccardSimilarity | GroupRank | filePair                                                                             |
|-------------------|-------------------------------|-----------|--------------------------------------------------------------------------------------|
| 0                 | java↔java                     | 1.000000  | 1AxonTimeLimitedTask↔UnitOfWorkTimeoutInterceptor                                    |
| 1                 | java↔java                     | 0.800000  | 2DoNotEnqueueTest↔ShouldEnqueueTest                                                  |
| 2                 | java↔java                     | 0.750000  | 3DefaultEventProcessorSpanFactory↔DefaultEventProcessorSpanFactoryTest               |
| 3                 | java↔java                     | 0.714286  | 4AvroUtil↔SpecificRecordBaseSerializerStrategy                                       |
| 4                 | java↔java                     | 0.700000  | 5DoNotEnqueue↔ShouldEnqueue                                                          |
| 5                 | java↔java                     | 0.625000  | 6AvroSerializerStrategy↔ByteArrayToGenericRecordConverter                            |
| 6                 | java↔java                     | 0.600000  | 7AxonTimeLimitedTask↔TimeoutWrappedMessageHandlingMember                             |
| 7                 | java↔java                     | 0.555556  | 8MessageAuthorizationDispatchInterceptor↔MessageAuthorizationDispatchInterceptorTest |
| 8                 | java↔java                     | 0.545455  | 9ShouldEnqueue↔DoNotEnqueueTest                                                      |
| 9                 | java↔java                     | 0.529412  | 10AnnotatedChildEntity↔ChildEntity                                                   |
| 10                | java↔xml                      | 0.180000  | 1JobRunrEventSchedulerTest↔pom                                                       |
| 11                | java↔xml                      | 0.156250  | 2ContextAddingInterceptor↔logback                                                    |
| 12                | java↔xml                      | 0.137405  | 3JobrunrDeadlineManagerTest↔pom                                                      |
| 13                | java↔xml                      | 0.134454  | 4AbstractDeadlineManagerTestSuite↔pom                                                |
| 14                | java↔xml                      | 0.130435  | 5JobRunrEventSchedulerTest↔pom                                                       |
| 15                | java↔xml                      | 0.113924  | 6JobRunrEventSchedulerTest↔pom                                                       |
| 16                | java↔xml                      | 0.111111  | 7TestAggregateRoot↔benchmark-mongo-context                                           |
| 17                | java↔xml                      | 0.109375  | 8Message↔benchmark-mongo-context                                                     |
| 18                | java↔xml                      | 0.108974  | 9AbstractDeadlineManagerTestSuite↔pom                                                |
| 19                | java↔xml                      | 0.106509  | 10JobrunrDeadlineManagerTest↔pom                                                     |
| 20                | java↔yaml                     | 0.220339  | 1JobRunrEventSchedulerTest↔antora                                                    |
| 21                | java↔yaml                     | 0.177419  | 2AvroSerializerAutoConfiguration↔antora                                              |
| 22                | java↔yaml                     | 0.174419  | 3JobRunrEventSchedulerTest↔slack-release-notification                                |
| 23                | java↔yaml                     | 0.171053  | 4AbstractDeadlineManagerTestSuite↔antora                                             |
| 24                | java↔yaml                     | 0.145631  | 5AbstractDeadlineManagerTestSuite↔slack-release-notification                         |
| 25                | java↔yaml                     | 0.144444  | 6JobrunrDeadlineManagerTest↔antora                                                   |
| 26                | java↔yaml                     | 0.128205  | 7JobrunrDeadlineManagerTest↔slack-release-notification                               |
| 27                | java↔yaml                     | 0.090909  | 8AbstractResourceInjector↔add-to-project                                             |
| 28                | java↔yaml                     | 0.081081  | 9ConfigurationScopeAwareProvider↔add-to-project                                      |
| 29                | java↔yaml                     | 0.071429  | 10AnnotatedMessageHandlingMemberTest↔antora                                          |
| 30                | adoc↔xml                      | 0.059322  | 1minor-releases↔pom                                                                  |
| 31                | adoc↔xml                      | 0.053571  | 2nav↔pom                                                                             |
| 32                | adoc↔xml                      | 0.051282  | 3rdbms-tuning↔pom                                                                    |
| 33                | adoc↔xml                      | 0.044872  | 4minor-releases↔pom                                                                  |
| 34                | adoc↔xml                      | 0.040000  | 5nav↔pom                                                                             |
| 35                | adoc↔xml                      | 0.039773  | 6minor-releases↔pom                                                                  |
| 36                | adoc↔xml                      | 0.038710  | 7rdbms-tuning↔pom                                                                    |
| 37                | adoc↔xml                      | 0.036232  | 8rdbms-tuning↔pom                                                                    |
| 38                | adoc↔xml                      | 0.036036  | 9event-snapshots↔pom                                                                 |
| 39                | adoc↔xml                      | 0.035294  | 10nav↔pom                                                                            |

## Co-Changed Files by their commit jaccard similarity



## Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

|    | fileExtensionPair | filePair                                                                 | combinedMetricsScore | updateCommitCountExter |
|----|-------------------|--------------------------------------------------------------------------|----------------------|------------------------|
| 0  | adoc↔xml          | index↔pom                                                                | 24                   |                        |
| 1  | adoc↔xml          | nav↔pom                                                                  | 27                   |                        |
| 2  | adoc↔xml          | nav↔pom                                                                  | 28                   |                        |
| 3  | adoc↔xml          | configuration↔pom                                                        | 28                   |                        |
| 4  | adoc↔xml          | minor-releases↔pom                                                       | 29                   |                        |
| 5  | adoc↔xml          | rdbms-tuning↔pom                                                         | 35                   |                        |
| 6  | adoc↔xml          | index↔pom                                                                | 38                   |                        |
| 7  | adoc↔xml          | nav↔pom                                                                  | 39                   |                        |
| 8  | adoc↔xml          | rdbms-tuning↔pom                                                         | 41                   |                        |
| 9  | adoc↔xml          | minor-releases↔pom                                                       | 41                   |                        |
| 10 | java↔java         | TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeoutInterceptorTest | 83                   |                        |
| 11 | java↔java         | AvroSchemaPackages↔AvroSchemaScan                                        | 83                   |                        |
| 12 | java↔java         | AvroSchemaScan↔ClasspathAvroSchemaLoader                                 | 83                   |                        |
| 13 | java↔java         | AvroSchemaPackages↔ClasspathAvroSchemaLoader                             | 83                   |                        |
| 14 | java↔java         | MetadataContextGetter↔MetadataContextSetter                              | 83                   |                        |
| 15 | java↔java         | DefaultEventProcessorSpanFactory↔DefaultEventProcessorSpanFactoryTest    | 86                   |                        |
| 16 | java↔java         | NoSuchDeadLetterException↔IgnoreTest                                     | 86                   |                        |
| 17 | java↔java         | SpanUtils↔SpanUtilsTest                                                  | 86                   |                        |
| 18 | java↔java         | DoNotEnqueueTest↔ShouldEnqueueTest                                       | 92                   |                        |
| 19 | java↔java         | AvroSerializerStrategy↔ByteArrayToGenericRecordConverter                 | 99                   |                        |
| 20 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 36                   |                        |
| 21 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 44                   |                        |
| 22 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 47                   |                        |
| 23 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 62                   |                        |
| 24 | java↔xml          | AbstractDeadlineManagerTestSuite↔pom                                     | 72                   |                        |
| 25 | java↔xml          | ContextAddingInterceptor↔logback                                         | 74                   |                        |
| 26 | java↔xml          | TokenAddingInterceptor↔logback                                           | 74                   |                        |
| 27 | java↔xml          | JobRunrEventSchedulerTest↔pom                                            | 80                   |                        |
| 28 | java↔xml          | JobRunrDeadlineManagerTest↔pom                                           | 84                   |                        |
| 29 | java↔xml          | UpstreamAwareStreamObserver↔logback                                      | 84                   |                        |
| 30 | java↔yaml         | JobRunrEventSchedulerTest↔antora                                         | 8                    |                        |
| 31 | java↔yaml         | JobRunrEventSchedulerTest↔slack-release-notification                     | 11                   |                        |
| 32 | java↔yaml         | AvroSerializerAutoConfiguration↔antora                                   | 14                   |                        |
| 33 | java↔yaml         | AbstractDeadlineManagerTestSuite↔slack-release-notification              | 22                   |                        |
| 34 | java↔yaml         | AbstractDeadlineManagerTestSuite↔antora                                  | 23                   |                        |
| 35 | java↔yaml         | JobRunrDeadlineManagerTest↔antora                                        | 27                   |                        |
| 36 | java↔yaml         | AxonTimeLimitedTaskTest↔antora                                           | 33                   |                        |
| 37 | java↔yaml         | JobRunrDeadlineManagerTest↔slack-release-notification                    | 33                   |                        |
| 38 | java↔yaml         | JobRunrEventSchedulerTest↔pullrequest                                    | 34                   |                        |
| 39 | java↔yaml         | JobRunrEventSchedulerTest↔main                                           | 36                   |                        |

## Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

Pairwise changed files with pareto-optimal metrics - not considering file extensions

|    | filePair                                                                                              | combinedMetricsScore | updateCommitCount |
|----|-------------------------------------------------------------------------------------------------------|----------------------|-------------------|
| 0  | AxonTimeLimitedTask↔UnitOfWorkTimeoutInterceptor                                                      | 100                  | 1                 |
| 1  | TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeoutInterceptorTest                              | 83                   | 1                 |
| 2  | EventProcessingModule↔EventProcessingModuleTest                                                       | 2786                 | 10                |
| 3  | AvroSchemaPackages↔AvroSchemaScan                                                                     | 83                   | 1                 |
| 4  | AvroSchemaScan↔ClasspathAvroSchemaLoader                                                              | 83                   | 1                 |
| 5  | AvroSchemaPackages↔ClasspathAvroSchemaLoader                                                          | 83                   | 1                 |
| 6  | JobRunrDeadlineManagerTest↔JobRunrEventSchedulerTest                                                  | 340                  | 2                 |
| 7  | JobRunrEventSchedulerTest↔AbstractDeadlineManagerTestSuite                                            | 345                  | 1                 |
| 8  | AxonServerQueryBus↔AxonServerCommandBus                                                               | 3563                 | 16                |
| 9  | SimpleEventHandlerInvoker↔SimpleEventHandlerInvokerTest                                               | 888                  | 4                 |
| 10 | DeadLetteringEventHandlerInvoker↔DeadLetteringEventHandlerInvokerTest                                 | 627                  | 2                 |
| 11 | JpaAutoConfiguration↔JpajavaxAutoConfiguration                                                        | 736                  | 2                 |
| 12 | JpaAutoConfigurationTest↔JpajavaxAutoConfigurationTest                                                | 301                  | 1                 |
| 13 | EventProcessingModule↔EventProcessingConfiguration                                                    | 2323                 | 3                 |
| 14 | EventPublicationOrderTest↔SynchronousLoopbackTest                                                     | 1574                 | 8                 |
| 15 | DoNotEnqueueTest↔ShouldEnqueueTest                                                                    | 92                   | 1                 |
| 16 | ReflectionUtilsTest↔ReflectionUtils                                                                   | 947                  | 2                 |
| 17 | MultiSourceTrackingTokenTest↔MultiSourceTrackingToken                                                 | 951                  | 2                 |
| 18 | SpringAxonAutoConfigurer↔AxonConfiguration                                                            | 2890                 | 10                |
| 19 | ResultValidatorImpl↔ResultValidator                                                                   | 1680                 | 9                 |
| 20 | FixtureExecutionResultImpl↔FixtureExecutionResult                                                     | 1316                 | 6                 |
| 21 | MetadataContextGetter↔MetadataContextSetter                                                           | 83                   | 1                 |
| 22 | AggregateMemberAnnotatedChildEntityCollectionDefinition↔AggregateMemberAnnotatedChildEntityDefinition | 290                  | 2                 |
| 23 | AggregateMemberAnnotatedChildEntityDefinition↔AggregateMemberAnnotatedChildEntityMapDefinition        | 293                  | 1                 |
| 24 | StubAggregate↔StubAggregate                                                                           | 836                  | 2                 |

Pairwise changed files with pareto-optimal metrics - using ranks grouped by file extensions

|   | fileExtensionPair | filePair                                    | combinedMetricsScore | updateCommitCount | updateCommitMinConfidence | updateCommitCount |
|---|-------------------|---------------------------------------------|----------------------|-------------------|---------------------------|-------------------|
| 0 | java↔java         | AggregateTestFixture↔TrackingEventProcessor | 14781                | 3                 | 0.007426                  |                   |

## WordCloud of git authors

|   | word                | frequency |
|---|---------------------|-----------|
| 0 | Steven van Beelen   | 3547      |
| 1 | Allard Buijze       | 2724      |
| 2 | smcvb               | 822       |
| 3 | Milan Savic         | 449       |
| 4 | Mitchell Herrijgers | 327       |
| 5 | Gerard Klijs        | 246       |
| 6 | Marc Gathier        | 231       |
| 7 | sara pellegrini     | 163       |
| 8 | Rene de Waele       | 121       |
| 9 | Jetro Coenradie     | 117       |

Wordcloud of git authors

